

Peter Hanukaev

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 0009-0006-0056-3091 DATE OF BIRTH: 1997-04-11 NATIONALITY: German

Education

01-2022 – present	Doctor of Philosophy in Computer Science Advisor: Harley Eades III Thesis: Generalized Graded Modal Type Theories and their Categorical Semantics (in preparation)	Augusta University, Augusta, GA, USA
02-2019 – 12-2021	Master of Science in Mathematics Thesis: Models of Type Theory via Representable Map Categories	University of Regensburg, Regensburg, Germany
10-2015 – 02-2019	Bachelor of Science in Mathematics Thesis: Classification of Vector Bundles	University of Regensburg, Regensburg, Germany

Research

Research Interests

Programming Languages | Formal Methods | Substructural Logic | Type Theory | Categorical Semantics

Granule Project

The Granule project is a research project with the goal of capturing information about program behavior at the type-system level, primarily using *graded* and *linear types*. Within the Granule project, I am focusing on developing a type system capable of tracking programs with respect to multiple grading disciplines at once, and describing the categorical semantics of such a type system.

Relevant Publications

1. Hanukaev, P. and Eades III, H. 2026. A unification of graded and substructural logics. Proceedings of the 42nd Conference on the Mathematical Foundations of Programming Semantics.
To appear in *Electronic Notes in Theoretical and Applied Informatics and Computer Science*.
Preproceedings: <https://ul-fmf.github.io/mfps-sstt-2026/files/pdfs/mfps/MFPS26-4.pdf>
2. Hanukaev, P. and Eades III, H. 2023. Combining Dependency, Grades, and Adjoint Logic. Proceedings of the 8th ACM SIGPLAN International Workshop on Type-Driven Development. ACM.
DOI:10.1145/3609027.3609408.

Employment

01-2022 – present	Graduate Research Assistant Research Group of Harley Eades III	Augusta University, Augusta, GA, USA
10-2017 – 12-2021	Teaching Assistant	University of Regensburg, Regensburg, Germany

Service

Peer Review

05-2026 Additional Reviewer – International Conference on Reversible Computation, RC 2026

Conference Organization

10-2023 Organizing Committee Member – Southeast Regional Programming Languages Seminar, SERPL 2023

University Service

05-2023 – 05-2026 President and co-founder of the $\Delta\Delta\Delta$ student organization, *Augusta University*
01-2023 – 05-2026 Co-organizer of $\Delta\Delta\Delta$'s weekly programming languages reading group, *Augusta University*

Teaching

10-2017 – 12-2021	Teaching Assistant for the following courses at <i>University of Regensburg</i> : • Linear Algebra I (10-2017 – 02-2018 and 10-2018 – 02-2019) • Linear Algebra II (04-2019 – 07-2019 and 04-2020 – 07-2020 and 04-2021 – 07-2021) • Algebra (10-2019 – 02-2020 and 10-2020 – 02-2021) • Analysis on Manifolds (04-2018 – 07-2018) • Introduction to Programming in Python (10-2021 – 12-2021)
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